Returns Root Cause Analysis & Financial Sustainability Report (2021)

Strategic Executive Audit for Key Stakeholders — Lumina Threads

1. Executive Summary & Financial Leakage

During 2021, Lumina Threads experienced an unprecedented surge in return rates, creating substantial bottom-line leakage and escalating operational challenges. This targeted diagnostic report isolates the primary return drivers and uncovers the "deep root cause" to empower executive decision-making and restore profit margins.

$59.3K	$1,983	26.54%
Lost Profit due to Returns	Reverse Logistics Wasted Cost	Overall Baseline Churn Rate

Direct Financial Impact: Product returns caused an immediate hit of **$59.3K** in Lost Profit and wasted an additional **$1,983** in Reverse Logistics Costs. Both metrics exhibit an alarming upward trend compared to the previous fiscal year.

2. Returns Behavioral Analysis (Category & Size)

A) Category Defects:

- **Bras & Tops:** Identified as a critical risk line, registering the company's highest category return rate at **10.49%**, corresponding to a direct loss of **$4K**.
- **Hoodies & Sweatshirts:** Outlined as the single heaviest absolute financial drain, leaking **$17.5K** in profit across **932** returned units due to higher retail price points.

B) Sizing Risk Concentrations:

Returns are heavily concentrated at extreme dimensional boundaries:

- **Size S:** Represents the highest absolute loss volume offender, with **634** units returned, resulting in **$14.3K** lost.
- **Size XL:** Presents the highest baseline product risk with a severe **7.23%** return rate, leaking **$13.8K**.

3. Product Category Deep Root Causes (The Why)

Advanced statistical modeling isolates specific independent variables that heavily multiply return probabilities:

- **Category Risk Multipliers:** The ***Bras & Tops*** line is the primary driver of returns, increasing the likelihood of a return by ***1.68×***, followed closely by ***Tank Tops*** at ***1.28×*** compared to other products.
- **Sizing Risk Multipliers:** A deep dimensional review confirms that size ***XL*** increases the overall return probability by ***1.16×***, strongly indicating a systematic fit or pattern measurement defect unique to this size configuration.

> 🚨 **The Critical Defect Segment**
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> By cross-referencing multiple drivers, the data isolates a highly concentrated risk segment where the return rate explodes to **14.7%**—more than double the company's baseline average. This critical segment is entirely concentrated in **Bras & Tops** when the product color is **Black**, single-handedly accounting for **92 returned units**.

4. Color Defects & Seasonal Spikes

A) Colorways Quality Control:

- **Lavender & White:** Represent severe quality control failures, registering the absolute highest individual return rates at **13%** ($474 lost) and **10%** ($3.6K lost) respectively.
- **Blue:** Acts as the primary macro volume bleeder, leaking **$11K** across **690** returned units due to its heavy baseline sales volume.

B) Seasonal & Timing Shocks:

The timeline reveals distinct operational shocks, with returns peaking critically in **August (Month 8)** and **October (Month 10)** at an alarming **10%** return rate each. These two peak months combined destroyed **$19.3K** in total profit.

5. Geographical Anomalies & The Deep Root Cause

On a national scale, England drives absolute loss volume at **$49K** lost. However, Scotland triggers the highest baseline regional risk with a steady **10%** return rate (**$9.3K** lost).

🛑 Supply Chain Breach: The XS/White Batch Anomaly (Dunbartonshire)

Inside Scotland, the region of **Dunbartonshire** presented a catastrophic operational outlier, yielding an unprecedented **47% return rate** across 44 units.

The Deep Root Cause: Filtering data specifically for size **XS** isolates a devastating **100% return rate correlation** (44 units total) exclusively tied to the **White** colorway within the **Hoodies & Sweatshirts** line.

⚠️ Downstream Strategic Implications:

1. **Historical Proof:** Shifting the timeline filter to 2020 completely empties the visual database for this defect, proving conclusively that this supply chain bug did not exist prior to 2021 and represents a new vendor/batch failure.
2. **New Customer Retention Failure:** This immediate physical defect severely damages first impressions; first-time buyers experiencing this batch error are completely discouraged from executing repeat orders.
3. **Scotland Loyalty Burn:** Cross-sector customer tracking confirms a critical post-return behavior where the majority of affected customers completely churned and never executed a second transaction, permanently damaging brand equity in Scotland.

6. Strategic Recommendations Matrix for Stakeholders

Target Domain	Immediate Corrective Action	Expected Business Impact
Supply Chain & Logistics	Halt and recall the Hoodies & Sweatshirts (Size XS, White) batch immediately; audit the Dunbartonshire supplier.	Eliminate the 47% local outlier and stop immediate reverse logistics waste.
Product Engineering	Re-calibrate dimensions for Size XL and investigate fabric/dye issues for Black Bras & Tops (14.7% defect segment).	Recover over $4K in lost margins and mitigate the 1.68× return multiplier.
Quality Control (QC)	Deploy strict pre-shipment inspection thresholds for Lavender and White colorways.	Reduce color-specific return baselines back to safe historical thresholds (<3%).
Customer Retention	Launch an automated retention campaign offering apologies and incentives to first-time buyers hit by the 2021 defects.	Reclaim Customer Lifetime Value (CLV) and repair the Scotland loyalty burn.